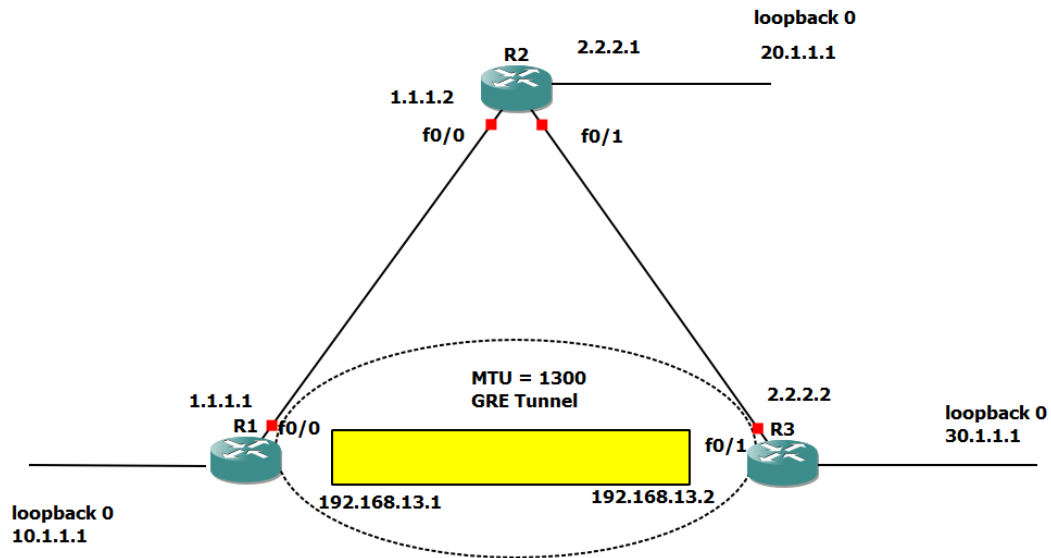


Practical No 3

Step 1: Design the Topology.



Step 2: Configure the System.

R1:

conf t

int

interface fastEthernet 0/0

ip address 1.1.1.1 255.0.0.0

no shut

exit

int

interface loopback 0

ip address 10.1.1.1 255.255.255.255

no shut

end

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip address 1.1.1.1 255.0.0.0
R1(config-if)#no shut
R1(config-if)#
*Sep 18 20:36:15.723: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Sep 18 20:36:16.723: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#interface loopback 0
R1(config-if)#
*Sep 18 20:36:29.187: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R1(config-if)#ip address 10.1.1.1 255.255.255.255
R1(config-if)#no shut
R1(config-if)#end
R1#
*Sep 18 20:36:51.443: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

R2:

conf t

hostname ISP

interface fastEthernet 0/0

ip address 1.1.1.2 255.0.0.0

no shut

exit

interface fastEthernet 0/1

ip address 2.2.2.1 255.0.0.0

no shut

exit

interface loopback 0

ip address 20.1.1.1 255.255.255.255

no shut

end

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#hostname ISP
ISP(config)#interface fastEthernet 0/0
ISP(config-if)#ip address 1.1.1.2 255.0.0.0
ISP(config-if)#no shut
ISP(config-if)#
*Sep 18 20:42:03.391: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Sep 18 20:42:04.391: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ISP(config-if)#exit
ISP(config)#interface fastEthernet 0/1
ISP(config-if)#ip address 2.2.2.1 255.0.0.0
ISP(config-if)#no shut
ISP(config-if)#
*Sep 18 20:42:39.603: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Sep 18 20:42:40.603: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ISP(config-if)#exit
ISP(config)#interface loopback 0
ISP(config-if)#
*Sep 18 20:42:49.607: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
ISP(config-if)#ip address 20.1.1.1 255.255.255.255
ISP(config-if)#no shut
ISP(config-if)#end
ISP#
*Sep 18 20:43:18.507: %SYS-5-CONFIG_I: Configured from console by console
ISP#
```

R3:

conf t

int

interface fa

interface fastEthernet 0/1

ip address 2.2.2.2 255.0.0.0

no shut

exit

```
interface loopback 0

ip address 30.1.1.1 255.255.255.255

no shut

end
```

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface fastEthernet 0/1
R3(config-if)#ip address 2.2.2.2 255.0.0.0
R3(config-if)#no shut
R3(config-if)#
*Sep 18 20:46:49.771: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Sep 18 20:46:50.771: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R3(config-if)#exit
R3(config)#interface loopback
% Incomplete command.

R3(config)#interface loopback 0
R3(config-if)#
*Sep 18 20:47:28.827: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R3(config-if)#ip address 30.1.1.1 255.255.255.255
R3(config-if)#no shut
R3(config-if)#end
R3#
*Sep 18 20:47:48.831: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

Step 3: Check the connection between R1, ISP, R3.

ISP#ping 1.1.1.1

```
ISP#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 28/31/32 ms
ISP#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 24/38/68 ms
ISP#
```

Step 4: Create the GRE Tunnel.

R1:

```
conf t

interface tunnel 1

tunnel source fastEthernet 0/0

tunnel destination 2.2.2.2

ip address 192.168.13.1 255.255.255.0

no shut

end

sh ip int bri
```

```

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface tunnel 1
R1(config-if)#
*Sep 18 20:56:49.855: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to down
R1(config-if)#tunnel source fastEthernet 0/0
R1(config-if)#tunnel destination 2.2.2.2
R1(config-if)#ip address 192.168.13.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#end
R1#
*Sep 18 20:58:49.371: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int br
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          1.1.1.1         YES manual up              up
FastEthernet0/1          unassigned      YES unset  administratively down down
GigabitEthernet1/0       unassigned      YES unset  administratively down down
Loopback0                10.1.1.1        YES manual up              up
Tunnel1                  192.168.13.1    YES manual up              down
R1#

```

R3:

```

conf t

interface tunnel 1

tunnel source fastEthernet 0/1

tunnel destination 1.1.1.1

ip address 192.168.13.2 255.255.255.0

no shut

end

sh ip int bri

```

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface tunnel 1
R3(config-if)#
*Sep 18 21:08:08.307: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to down
R3(config-if)#tunnel source fastEthernet 0/1
R3(config-if)#tunnel destination 1.1.1.1
R3(config-if)#ip address 192.168.13.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#end
R3#
*Sep 18 21:10:14.923: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip int bri
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          unassigned      YES unset  administratively down down
FastEthernet0/1          2.2.2.2         YES manual up              up
GigabitEthernet1/0       unassigned      YES unset  administratively down down
Loopback0                30.1.1.1        YES manual up              up
Tunnel1                  192.168.13.2    YES manual up              down
R3#

```

Step 5: Assign a Static Route for R1 & R3.

R1:

```

conf t

ip route 2.0.0.0 255.0.0.0 1.1.1.2

end

```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 2.0.0.0 255.0.0.0 1.1.1.2
R1(config)#
*Sep 18 21:11:18.891: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up
R1(config)#end
R1#
*Sep 18 21:11:25.315: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

R3:

conf t

ip route 1.0.0.0 255.0.0.0 2.2.2.1

end

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 1.0.0.0 255.0.0.0 2.2.2.1
R3(config)#end
R3#
*Sep 18 21:12:02.883: %SYS-5-CONFIG_I: Configured from console by console
R3#
*Sep 18 21:12:07.347: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up
R3#
```

Step 6: Check whether the tunnel works.

R1#ping 192.168.13.2

```
R1#ping 192.168.13.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.13.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/65/76 ms
R1#
```

Step 7: Configure EIGRP for R1 & R3.

R1:

configure terminal

router eigrp 1

network 10.0.0.0

network 192.168.13.0

no auto-summary

end

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router eigrp 1
R1(config-router)#network 10.0.0.0
R1(config-router)#network 192.168.13.0
R1(config-router)#no auto-summary
R1(config-router)#end
R1#
*Sep 18 21:13:41.199: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

R3:

configure terminal

router eigrp 1

network 30.0.0.0

network 192.168.13.0

no auto-summary

end

```
R3(config)#router eigrp 1
R3(config-router)#network 30.0.0.0
R3(config-router)#network 192.168.13.0
R3(config-router)#no auto-summary
R3(config-router)#end
R3#
*Sep 18 21:14:13.451: %SYS-5-CONFIG_I: Configured from console by console
*Sep 18 21:14:13.543: %DUAL-5-NBRCHANGE: EIGRP-IPv4 1: Neighbor 192.168.13.1 (Tunnel1) is up: new adjacency
R3#
```

Step 8: Check whether EIGRP is configured for R1 & R3.

• **R1:**

R1#sh ip route

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       1.0.0.0/8 is directly connected, FastEthernet0/0
L       1.1.1.1/32 is directly connected, FastEthernet0/0
S       2.0.0.0/8 [1/0] via 1.1.1.2
        10.0.0.0/32 is subnetted, 1 subnets
C       10.1.1.1 is directly connected, Loopback0
        30.0.0.0/32 is subnetted, 1 subnets
D       30.1.1.1 [90/27008000] via 192.168.13.2, 00:00:31, Tunnel1
        192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/24 is directly connected, Tunnel1
L       192.168.13.1/32 is directly connected, Tunnel1
R1#
```

R3:

R3#sh ip route

```
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override
```

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 2.2.2.1
    2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    2.0.0.0/8 is directly connected, FastEthernet0/1
L    2.2.2.2/32 is directly connected, FastEthernet0/1
    10.0.0.0/32 is subnetted, 1 subnets
D    10.1.1.1 [90/27008000] via 192.168.13.1, 00:01:36, Tunnel1
    30.0.0.0/32 is subnetted, 1 subnets
C    30.1.1.1 is directly connected, Loopback0
    192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/24 is directly connected, Tunnel1
L    192.168.13.2/32 is directly connected, Tunnel1
R3#
```

Step 9: To set MTU as the GRE Head.

R1:

configure terminal

interface tunnel 1

ip mtu 1300

ip tcp adjust-mss 1360

end

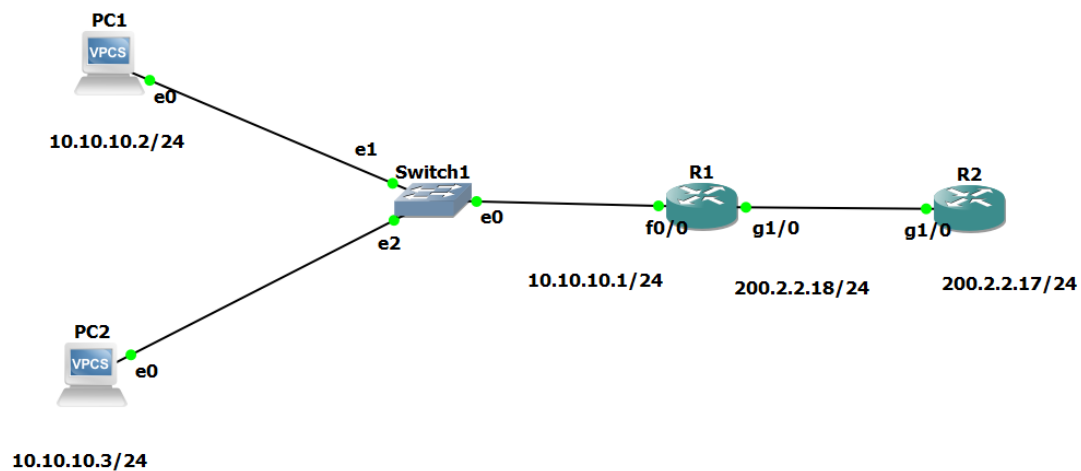
```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface tunnel 1
R1(config-if)#ip mtu 1300
R1(config-if)#ip tcp adjust-mss 1360
R1(config-if)#end
R1#
*Sep 18 21:16:40.315: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

2. Implement NAT:

What is NAT?

Network Address Translation (NAT) is a process in which one or more local IP address is translated into one or more Global IP address and vice versa in order to provide Internet access to the local hosts. Also, it does the translation of port numbers i.e. masks the port number of the host with another port number, in the packet that will be routed to the destination. It then makes the corresponding entries of IP address and port number in the NAT table. NAT generally operates on a router or firewall.

Step 1: Design the Topology. Use c7200 Router IOS



Step 2: Configure the network.

R1(gateway):

configure terminal

hostname gateway

interface GigabitEthernet 1/0

ip address 200.2.2.18 255.255.255.252

no shutdown

interface fastEthernet 0/0

ip address 10.10.10.1 255.255.255.0

no shutdown

end


```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname gateway
gateway(config)#int g1/0
gateway(config-if)#ip address 200.2.2.18 255.255.255.252
gateway(config-if)#no shutdown
gateway(config-if)#
gateway(config-if)#
*Sep 20 17:09:04.007: %LINK-3-UPDOWN: Interface GigabitEthernet1/0, changed state to up
*Sep 20 17:09:05.007: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0, changed state to up
gateway(config-if)#interface fastEthernet 0/0
gateway(config-if)#ip address 10.10.10.1 255.255.255.0
gateway(config-if)#no shutdown
gateway(config-if)#end
gateway#
*Sep 20 17:09:30.559: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Sep 20 17:09:31.559: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
gateway#
*Sep 20 17:09:32.059: %SYS-5-CONFIG_I: Configured from console by console
gateway#
```

R2(ISP):

configure terminal

hostname ISP

interface loopback0

ip address 172.16.1.1 255.255.255.255

no shutdown

interface GigabitEthernet1/0

ip address 200.2.2.17 255.255.255.252

no shutdown

end

```
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#hostname ISP
ISP(config)#interface loopback0
ISP(config-if)#
*Sep 20 17:11:49.627: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
ISP(config-if)#ip address 172.16.1.1 255.255.255.255
ISP(config-if)#no shutdown
ISP(config-if)#
ISP(config-if)#interface GigabitEthernet1/0
ISP(config-if)#ip address 200.2.2.17 255.255.255.252
ISP(config-if)#no shutdown
ISP(config-if)#end
ISP#
*Sep 20 17:12:27.207: %LINK-3-UPDOWN: Interface GigabitEthernet1/0, changed state to up
*Sep 20 17:12:28.207: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0, changed state to up
ISP#
*Sep 20 17:12:28.511: %SYS-5-CONFIG_I: Configured from console by console
ISP#
```

PC1:

PC1> ip 10.10.10.2 255.255.255.0 10.10.10.1

PC1> sh ip

```
PC1> ip 10.10.10.2 255.255.255.0 10.10.10.1
Checking for duplicate address...
PC1 : 10.10.10.2 255.255.255.0 gateway 10.10.10.1

PC1> sh ip

NAME          : PC1[1]
IP/MASK        : 10.10.10.2/24
GATEWAY        : 10.10.10.1
DNS            :
MAC           : 00:50:79:66:68:00
LPORT         : 10014
RHOST:PORT    : 127.0.0.1:10015
MTU:          : 1500

PC1> █
```

PC2:

PC2> ip 10.10.10.3 255.255.255.0 10.10.10.1

PC2> sh ip

```
PC2> ip 10.10.10.3 255.255.255.0 10.10.10.1
Checking for duplicate address...
PC1 : 10.10.10.3 255.255.255.0 gateway 10.10.10.1

PC2> sh ip

NAME          : PC2[1]
IP/MASK        : 10.10.10.3/24
GATEWAY        : 10.10.10.1
DNS            :
MAC           : 00:50:79:66:68:01
LPORT         : 10016
RHOST:PORT    : 127.0.0.1:10017
MTU:          : 1500

PC2> █
```

Step 3: Create a Static Route.

R2:

ISP#conf t

ISP (config)#ip route 199.99.9.32 255.255.255.224 200.2.2.18

ISP(config)#end

```
ISP#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ISP(config)#ip route 199.99.9.32 255.255.255.224 200.2.2.18
ISP(config)#end
ISP#
*Sep 20 17:16:47.179: %SYS-5-CONFIG_I: Configured from console by console
ISP#
```

Step 4: Create a Default Route.

```
gateway#conf t
```

```
gateway (config)#ip route 0.0.0.0 0.0.0.0 200.2.2.17
```

```
gateway#conf t
Enter configuration commands, one per line. End with CNTL/Z.
gateway(config)#ip route 0.0.0.0 0.0.0.0 200.2.2.17
```

Step 5: Make a pool of IP Address which can be used as Public IP Address:

Run the command: **ip nat pool public-access 199.99.9.32 199.99.9.35 netmask 255.255.255.252**

configure terminal

```
ip nat pool public-access 199.99.9.32 199.99.9.46 netmask 255.255.255.240
```

end

```
gateway(config)#access 199.99.9.32 199.99.9.46 netmask 255.255.255.240
gateway(config)#no access-list 1
gateway(config)#access-list 1 permit 10.10.10.0 0.0.0.255
gateway(config)#ip nat inside source list 1 pool public-access overload
gateway(config)#interface fastEthernet0/0
gateway(config-if)# ip nat inside
gateway(config-if)#interface g1/0
gateway(config-if)#ip nat outside
gateway(config-if)#end
```

Step 6: Make an access list that will map the public IP addresses to the inside private IP addresses and define the NAT translation from inside list to outside pool.

```
gateway (config)#access-list 1 permit 10.10.10.0 0.0.0.255
```

```
gateway(config)#access-list 1 permit 10.10.10.0 0.0.0.255
```

Step 7: Now we will define which interface is inside and which one is outside.

Gateway:

```
ip nat inside source list 1 pool public-access overload
```

```
interface fastEthernet 0/0
```

```
ip nat inside
```

```
interface g 1/0
```

```
ip nat outside
```

```
end
```

```
gateway(config)#access-list 1 permit 10.10.10.0 0.0.0.255
gateway(config)#ip nat inside source list 1 pool public-access overload
gateway(config)#interface fastEthernet0/0
gateway(config-if)# ip nat inside
gateway(config-if)#interface g1/0
gateway(config-if)#ip nat outside
gateway(config-if)#end
gateway#
```

Step 8: Now ping from the PC to the loopback of ISP.

PC1:

```
PC1> ping 172.16.1.1
```

```
PC1>
PC1> ping 172.16.1.1
84 bytes from 172.16.1.1 icmp_seq=1 ttl=254 time=46.471 ms
84 bytes from 172.16.1.1 icmp_seq=2 ttl=254 time=45.639 ms
84 bytes from 172.16.1.1 icmp_seq=3 ttl=254 time=45.762 ms
84 bytes from 172.16.1.1 icmp_seq=4 ttl=254 time=46.062 ms
84 bytes from 172.16.1.1 icmp_seq=5 ttl=254 time=45.432 ms
```

PC2:

```
PC2> ping 172.16.1.1
```

```
PC2> ping 172.16.1.1
84 bytes from 172.16.1.1 icmp_seq=1 ttl=254 time=46.042 ms
84 bytes from 172.16.1.1 icmp_seq=2 ttl=254 time=45.017 ms
84 bytes from 172.16.1.1 icmp_seq=3 ttl=254 time=47.011 ms
84 bytes from 172.16.1.1 icmp_seq=4 ttl=254 time=45.702 ms
84 bytes from 172.16.1.1 icmp_seq=5 ttl=254 time=46.748 ms
```

Step 9: Verify NAT & PAT Translations.

```
#show run | include nat
```

```
gateway#show run | include nat
ip nat inside
ip nat outside
ip nat pool public-access 199.99.9.32 199.99.9.46 netmask 255.255.255.240
ip nat inside source list 1 pool public-access overload
```

```
gateway#sh ip nat translations
```

```
gateway#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 199.99.9.33:1024  10.10.10.2:34972  172.16.1.1:34972  172.16.1.1:1024
icmp 199.99.9.33:1025  10.10.10.2:35228  172.16.1.1:35228  172.16.1.1:1025
icmp 199.99.9.33:1026  10.10.10.2:35484  172.16.1.1:35484  172.16.1.1:1026
icmp 199.99.9.33:1027  10.10.10.2:35740  172.16.1.1:35740  172.16.1.1:1027
icmp 199.99.9.33:1028  10.10.10.2:35996  172.16.1.1:35996  172.16.1.1:1028
icmp 199.99.9.33:1029  10.10.10.3:40604  172.16.1.1:40604  172.16.1.1:1029
icmp 199.99.9.33:1030  10.10.10.3:40860  172.16.1.1:40860  172.16.1.1:1030
icmp 199.99.9.33:1031  10.10.10.3:41116  172.16.1.1:41116  172.16.1.1:1031
icmp 199.99.9.33:1032  10.10.10.3:41372  172.16.1.1:41372  172.16.1.1:1032
icmp 199.99.9.33:1033  10.10.10.3:41628  172.16.1.1:41628  172.16.1.1:1033
```

Step 10: Verify NAT & PAT Statistics.

```
gateway#sh ip nat statistics
```

```
gateway#sh ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Outside interfaces:
  GigabitEthernet1/0
Inside interfaces:
  FastEthernet0/0
Hits: 50 Misses: 0
CEF Translated packets: 50, CEF Punted packets: 15
Expired translations: 25
Dynamic mappings:
-- Inside Source
[Id: 1] access-list 1 pool public-access refcount 0
  pool public-access: netmask 255.255.255.240
    start 199.99.9.32 end 199.99.9.46
    type generic, total addresses 15, allocated 0 (0%), misses 0
nat-limit statistics:
  max entry: max allowed 0, used 0, missed 0
gateway#
```

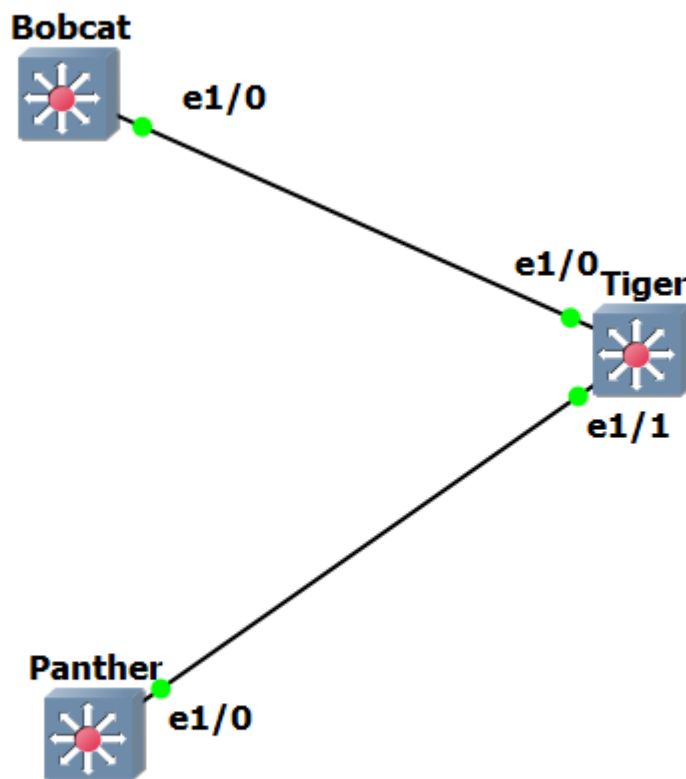
Implement VTP

Objective

Create the following VLANS on switch Bobcat:

- VLAN 10: name Tigers
- VLAN 20: name Lions
- VLAN 30: name Panthers
- Configure the interfaces between the switches as trunks.
- Configure switch Bobcat to be the VTP server.
- Configure switch Panther to be a VTP client.
- Configure switch Tiger so it does not synchronize itself to the latest VTP information, it should
forward advertisements to switch Panther though.
- Change the VTP domain name to "MSCCS".
- Use the password "MSCCS123" for VTP.
- Make sure there is no unnecessary vlan traffic flooded on the trunk links.

VTP Server



VTP Client

Create the following VLANS on switch Bobcat:

VLAN 10: name Tigers

VLAN 20: name Lions

VLAN 30: name Panthers

conf t

vlan 10

vlan 20

vlan 30

exit

end

show vlan brief

```
Bobcat#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Bobcat(config)#vlan 10
Bobcat(config-vlan)#vlan 20
Bobcat(config-vlan)#vlan 30
Bobcat(config-vlan)#exit
Bobcat(config)#end
Bobcat#
*Sep 21 09:46:04.240: %SYS-5-CONFIG_I: Configured from console by console
Bobcat#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/1, Et1/2, Et1/3, Et2/0 Et2/1, Et2/2, Et2/3, Et3/0 Et3/1, Et3/2, Et3/3
10	VLAN0010	active	
20	VLAN0020	active	
30	VLAN0030	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Bobcat#

Configure the interfaces between the switches as trunks.

Bobcat(config)#int range e1/0

Bobcat (config-if-range)#switchport trunk encapsulation dot1q

```
Bobcat(config)#int range e1/0
Bobcat(config-if-range)#switchport trunk encapsulation dot1q
Bobcat(config-if-range)#
```

Tiger#conf t

Tiger(config)#int range e1/0

Tiger (config-if-range)# switchport trunk encapsulation dot1q

Tiger (config-if-range)#exit

Tiger (config)#int range e1/1

Tiger (config-if-range)#switchport trunk encapsulation dot1q

Tiger (config-if-range)#exit

Tiger (config)#

```
Tiger#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Tiger(config)#int range e1/0
Tiger(config-if-range)#switchport trunk encapsulation dot1q
Tiger(config-if-range)#exit
Tiger(config)#int range e1/1
Tiger(config-if-range)#switchport trunk encapsulation dot1q
Tiger(config-if-range)#exit
Tiger(config)#
```

Panther#conf t

Panther (config)#int range e1/0

Panther (config-if-range)#switchport trunk encapsulation dot1q

Panther (config-if-range)#exit

```
Panther#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Panther(config)#int range e1/0
Panther(config-if-range)#switchport trunk encapsulation dot1q
Panther(config-if-range)#exit
Panther(config)#
```

Configure switch Bobcat to be the VTP server.

int e1/0

vtp mode server

```
Bobcat(config)#int e1/0
Bobcat(config-if)#vtp mode server
Device mode already VTP Server for VLANs.
Bobcat(config)#
```

Configure switch Panther to be a VTP client.

int e1/0

vtp mode client


```
Panther(config)#int e1/0
Panther(config-if)#vtp mode client
Setting device to VTP Client mode for VLANs.
Panther(config)#
```

Configure switch Tiger so it does not synchronise itself to the latest VTP information, it should forward advertisements to switch Panther though.

int e1/1

vtp mode transparent

```
Tiger(config)#int e1/1
Tiger(config-if)#vtp mode trans
Setting device to VTP Transparent mode for VLANs.
Tiger(config)#
```

Change the VTP domain name to "MSCCS".

Tiger

vtp domain MSCCS

```
Tiger(config)#vtp domain MSCCS
Changing VTP domain name from NULL to MSCCS
Tiger(config)#
```

Bobcat

vtp domain MSCCS

```
Bobcat(config)#vtp domain MSCCS
Changing VTP domain name from NULL to MSCCS
Bobcat(config)#
```

Panther

vtp domain MSCCS

```
Panther(config)#vtp domain MSCCS
Domain name already set to MSCCS.
Panther(config)#
```

Use the password "MSCCS123" for VTP.

Bobcat

vtp password MSCCS123

```
Bobcat(config)#vtp password MSCCS123
Setting device VTP password to MSCCS123
Bobcat(config)#
```

Tiger

vtp password MSCCS123

```
Tiger(config)#vtp password MSCCS123
Setting device VTP password to MSCCS123
Tiger(config)#
```

Panther

vtp password MSCCS123

```
Panther(config)#vtp password MSCCS123
Setting device VTP password to MSCCS123
```

Check and verify the VLANs

Bobcat#show vlan

```
Bobcat#sh vlan

VLAN Name                Status    Ports
-----
1    default                active    Et0/0, Et0/1, Et0/2, Et0/3
                                           Et1/1, Et1/2, Et1/3, Et2/0
                                           Et2/1, Et2/2, Et2/3, Et3/0
                                           Et3/1, Et3/2, Et3/3
10   VLAN0010                active
20   VLAN0020                active
30   VLAN0030                active
1002 fddi-default            act/unsup
1003 token-ring-default     act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1    enet   100001    1500  -      -      -      -   -        0      0
10   enet   100010    1500  -      -      -      -   -        0      0
20   enet   100020    1500  -      -      -      -   -        0      0
30   enet   100030    1500  -      -      -      -   -        0      0
1002 fddi   101002    1500  -      -      -      -   -        0      0
1003 tr    101003    1500  -      -      -      -   -        0      0
1004 fdnet 101004    1500  -      -      -      ieee -        0      0
1005 trnet 101005    1500  -      -      -      ibm  -        0      0

Primary Secondary Type      Ports
-----
```

Tiger#show vlan

Tiger#show vlan

VLAN Name					Status	Ports					

1	default				active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/2, Et1/3, Et2/0, Et2/1 Et2/2, Et2/3, Et3/0, Et3/1 Et3/2, Et3/3					
1002	fddi-default				act/unsup						
1003	token-ring-default				act/unsup						
1004	fddinet-default				act/unsup						
1005	trnet-default				act/unsup						
VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2	

1	enet	100001	1500	-	-	-	-	-	0	0	
1002	fddi	101002	1500	-	-	-	-	-	0	0	
1003	tr	101003	1500	-	-	-	-	-	0	0	
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0	
1005	trnet	101005	1500	-	-	-	ibm	-	0	0	
Primary Secondary Type					Ports						

Panther#show vlan

Panther#show vlan

VLAN Name					Status	Ports					

1	default				active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/1, Et1/2, Et1/3, Et2/0 Et2/1, Et2/2, Et2/3, Et3/0 Et3/1, Et3/2, Et3/3					
10	VLAN0010				active						
20	VLAN0020				active						
30	VLAN0030				active						
1002	fddi-default				act/unsup						
1003	token-ring-default				act/unsup						
1004	fddinet-default				act/unsup						
1005	trnet-default				act/unsup						
VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2	

1	enet	100001	1500	-	-	-	-	-	0	0	
10	enet	100010	1500	-	-	-	-	-	0	0	
20	enet	100020	1500	-	-	-	-	-	0	0	
30	enet	100030	1500	-	-	-	-	-	0	0	
1002	fddi	101002	1500	-	-	-	-	-	0	0	
1003	tr	101003	1500	-	-	-	-	srb	0	0	
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0	
1005	trnet	101005	1500	-	-	-	ibm	-	0	0	
Primary Secondary Type					Ports						

Panther#show vtp status

```

Panther#show vtp status
VTP Version capable      : 1 to 3
VTP version running      : 1
VTP Domain Name          : MSCCS
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : aabb.cc00.0300
Configuration last modified by 0.0.0.0 at 9-21-25 09:46:00

Feature VLAN:
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 1005
Number of existing VLANs : 8
Configuration Revision    : 3
MD5 digest               : 0xE1 0x62 0xBD 0x80 0x24 0xC3 0xFE 0x3F
                        : 0x2C 0xF9 0xEE 0xA0 0x6B 0x8B 0xC2 0x15

```

Make sure there is no unnecessary vlan traffic flooded on the trunk links.

Bobcat:

conf t

vtp pruning

```

Bobcat#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Bobcat(config)#vtp pruning
Pruning switched on
Bobcat(config)#

```

Panther#show vtp status

```

Panther#show vtp status
VTP Version capable      : 1 to 3
VTP version running      : 1
VTP Domain Name          : MSCCS
VTP Pruning Mode         : Enabled
VTP Traps Generation     : Disabled
Device ID                : aabb.cc00.0300
Configuration last modified by 0.0.0.0 at 9-21-25 10:06:06

Feature VLAN:
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 1005
Number of existing VLANs : 8
Configuration Revision    : 4
MD5 digest               : 0x1D 0x4F 0x8B 0x72 0x1E 0xF3 0xE8 0x9B
                        : 0x47 0x72 0x71 0x05 0x0D 0x77 0x18 0x60

```